Manifold for Type 1 Diabetes

Research-partnership brief — pre-clinical risk stratification and intervention-window detection

1 · THE CLINICAL PROBLEM

Type 1 Diabetes has a years-long pre-clinical prodrome:

- **Stage 1** — ≥2 persistent islet autoantibodies, normoglycemia.
- **Stage 2** — dysglycemia (abnormal OGTT), still asymptomatic.
- **Stage 3** — clinical onset (~100% 10-year risk conditional on Stage 1).

Current risk stratification tools (TrialNet DPTRS, Index60, M120) deliver strong population-level survival curves but miss **individual-trajectory inflection points** — the window where a Stage-2-appropriate intervention (teplizumab-class anti-CD3, low-dose ATG, experimental antigen-specific therapies) has maximal benefit.

What today's tools do not do well:

- Treat autoantibody seroconversion as richer than a binary count.
- Use continuous glycemic variability *before* dysglycemia as signal.
- Encode causal structure between HLA haplotype, viral exposure (enterovirus / Coxsackie B), gut microbiome, and metabolic load in a unified model rather than separate correlation studies.

> Teplizumab's 2022 approval made the pre-clinical window *actionable*, not academic. Every month of better per-patient stratification is intervention-window bought.

2 · WHAT MANIFOLD BRINGS

Manifold is a causal-intelligence platform for detecting structural fragility and regime transitions in complex multi-signal systems. Three engine capabilities anchor the T1D configuration:

1. **Causal discovery (Spirtes / PC algorithm) on multi-omic cohorts.** Replaces ad-hoc correlation practice in GWAS × microbiome × virome × metabolomic integration with a principled conditional-independence-based DAG.
2. **Criticality estimators feeding a composite fragility score (ΩF).** Estimators configured to the domain's dynamics; the composite is the stable integration layer.
3. **Pearl do-calculus / interdiction.** Frame therapy selection as max-attenuation intervention on the inferred causal graph — directly parallel to teplizumab-class "delay onset" trial design.

3 · ΩF PILLAR MAPPING – T1D PHYSIOLOGY

ΩF PILLAR	T1D READING
I — Interaction	Autoantibody-to-autoantibody coupling (IAA, GADA, IA-2A, ZnT8); HLA × viral-trigger epistasis
R — Reserve	β-cell functional mass proxied by C-peptide AUC on MMTT; glucose-response amplitude
J — Junction	Immune–metabolic subsystem coupling (transfer entropy between immune-cell subsets and glycemic variability)
C — Cascade	Autoreactive T-cell clonal expansion; epitope spreading rate
T — Temporal	Hazard of next stage transition given current trajectory

4 · CRITICALITY ESTIMATOR CONFIGURATION

T1D pre-clinical progression is a staged Markov / survival process with autoreactive positive feedback. The criticality layer is configured accordingly:

ESTIMATOR	WHAT IT CATCHES	PILLAR FEED
Dynamic Network Biomarkers (Chen et al. 2012)	Leading-subset variance + correlation spike *before* overt stage transition	I, T
Bayesian online change-point (BOCPD)	Discrete Stage 1→2→3 transition timing with calibrated uncertainty	T, R
Deep survival with time-varying covariates (DeepSurv / RNN-Surv)	Personalized hazard given longitudinal autoantibody + CGM + metabolic signals	C, T

Complementary estimators available in the same layer for v2:

- **Variance-based early-warning signals** on rolling CGM windows once continuous glucose data is onboarded — feeds Temporal.
- **Persistent homology** on longitudinal multi-omic point clouds for topological regime detection across integrated immune-metabolic trajectories — feeds Interaction and Cascade.

5 · MINIMUM VIABLE COLLABORATION (V1)

Smallest slice that produces something a principal investigator will take seriously.

- **Dataset.** TEDDY retrospective cohort slice — ~8,000 children, 15 years of serial autoantibody + HLA + limited metabolomic + environmental exposure data. TrialNet PTP as secondary validation.
- **Access path.** NIDDK Central Repository DUA, co-sponsored with a TEDDY-affiliated PI.
- **Deliverable.** Per-subject **Ω**F trajectory with DNB + change-point + deep-survival agreement, benchmarked against DPTRS on Stage-3 prediction at 1 / 3 / 5-year horizons.
- **Success criteria.** AUC lift ≥ 0.05 over DPTRS on matched subjects; calibration improvement (Brier score, calibration-in-the-large); human-interpretable **Ω**F-pillar decomposition per subject.
- **Timeline.** 10–12 weeks from data access.
- **Out of scope v1** (reserved for v2 / prospective arm): live CGM, gut microbiome sequencing, virome, fecal metabolomics.

6 · WHY THIS, WHY NOW

- **Teplizumab is approved.** The pre-clinical window is actionable. Stage-2 patients are candidates today.
- **TEDDY is mature.** Data access pathways are established; the analytic community is sophisticated and receptive.
- **Funding lines are open.** JDRF Prediction & Prevention, Helmsley Charitable Trust T1D program, NIH NIDDK SBIR/STTR, and the Breakthrough T1D (JDRF) Industry Discovery & Development Partnerships track all have active RFPs in this space.
- **The platform is ready.** Manifold's causal, criticality, and interdiction engines are production-grade; configuring them for the T1D domain is bounded, well-scoped work.

7 · ASK

Partnership with a TEDDY-affiliated PI (or equivalent cohort steward — TrialNet, DiPiS, BabyDiab) for:

1. Retrospective data access under their existing Data Use Agreement.
2. Clinical co-authorship on the benchmark paper.
3. Co-application to JDRF / Helmsley / NIH for the v2 prospective arm.

In exchange, Apex Analytica brings the causal-inference engine, the **Ω**F framework, and the software team. Apex retains commercial rights to the platform; research IP from the joint work is shared per standard academic-industry terms.